Alessandro Aimi

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Modena, Italy

Postdoctoral researcher in Computer Science and Engineering

Long-range low-power wireless IoT networks
Network simulation, modeling and optimization

Employment History

Research Fellow, **Oct. 2023 - Current**
Mobile Middleware Research Group, DISI,
Alma Mater Studiorum - Università di Bologna, Italy.

Researcher, **2020 - 2023**
5G Core Network Management & Automation Team, Orange Labs,
Orange SA, Paris, France.

Education History

PhD in Computer Science, **2020 - 2023**
Conservatoire national des arts et métiers, Paris, France.
Thesis title: Algorithms for Reliability Insurance and Traffic Differentiation in Low Power Internet of Things Communication Networks.
Supervisor: Prof. Stefano Secci.

MSc in Computer Science and Engineering, **2017 - 2020**
Politecnico di Milano, Italy.
Study track: Theoretical Computer Science and Artificial Intelligence.
Thesis title: Efficient Algorithms for Computing Semivalue Single Point Solution Concepts of TU Cooperative Games.
Supervisor: Prof. Roberto Lucchetti.

Collaborative Projects

House of Emerging Technologies, City of Bologna, **2023 - 2025**
Italian innovation project aimed at supporting local startups and enterprises in exploiting new technologies enabled by 5G networks, such as Blockchain, IoT, and AI.
Leadership of WP3 (Internal experimentation and development of PoCs with innovative technologies on the territory).

ANR INTELLIGENTSIA, **2020 - 2024**
French research project on INTelligent Edge using Learning Loops & Information GEneration for NeTwork State Inference-based Automation.
Full-time involvement up to Oct. 2023, only partial since then. Contribution to WP1 (Functional architecture framework definition), WP2 (Orchestration state machine modeling & learning algorithms), WP3 (Virtualization and automation of IoT access networks), and WP4 (Proof of Concept and Experimentation).

H2020 EU Flagship Hexa-X, **2020 - 2021**
European research project on B5G/6G vision and intelligent fabric of technology enablers connecting the human, physical, and digital worlds.
Participation to WP5 (Architectural enablers for B5G/6G), and WP6 (Intelligent orchestration and service management for future B5G/6G networks).

Technical Affiliations

Institute of Electrical and Electronics Engineers (IEEE) **2020 - Current**
Member

Technical Skills

LoRaWAN, C/C++, Rust, GNU/Linux, git, Docker, Python, MATLAB, SQL, Prometheus, InfluxDB, Grafana, OpenTelemetry, GitHub Actions

Languages

Italian, **English**, French